

CS383 - Software Engineering

Software Design (II)

Behaviour models

- Describe the dynamic behaviour of the system and its components when respond to
 - Data: incoming data that the system must process
 - Events: that happen because of actions that the system must process.

Why we care about Data?

- Data is very valuable asset nowadays
 - Most of big tech companies rely on data for user experience.
 - Most research and marketing rely on data and data science
- However, privacy and security of data have become a challenge for many
- Also, managing data is an important aspect in software engineering

Data-driven modeling

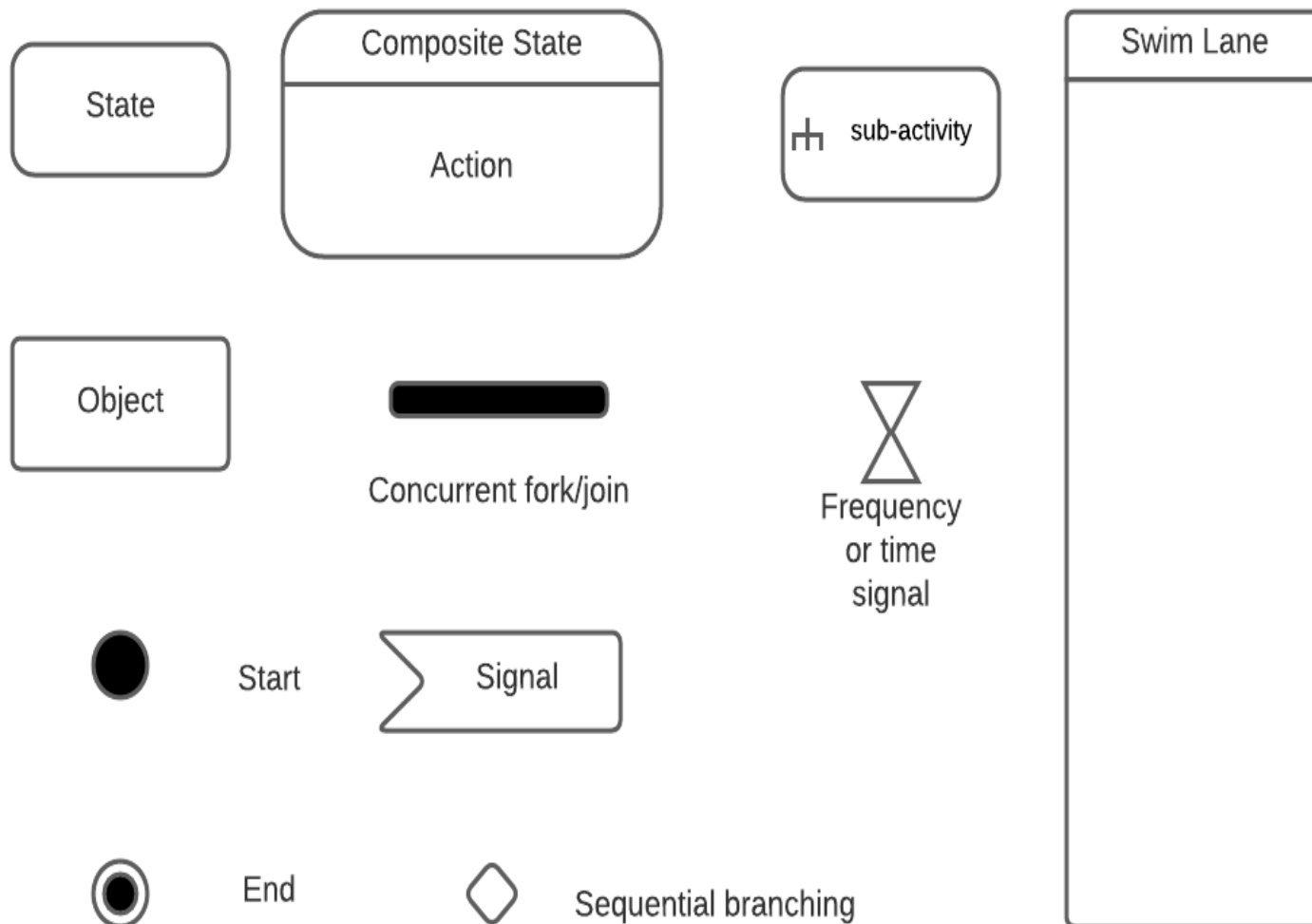
A modeling process that shows the sequence of actions involved in processing data and generating information or other data.

- Firstly introduced in 1978 by DeMarco
 - Introduced Data-Flow Diagrams to show processing steps of a system
 - Purely focus on data and how they are processed by system components
- UML doesn't have pure data modeling diagram
 - Data are modeled as part of interaction between objects
 - E.g. using sequence diagram, activity diagram
- For data-driven modeling we focus on
 - DFD, activity diagram

Activity diagram

- Describe processes of systems and its subsystems
- Ideal for use with workflows
- Show flow of control from one activity to another
- Activity diagrams show activities as nodes, with flow of control shown as arcs or edges
- A state in activity diagram:
 - has only one income and one outcome arc/edge
 - may be composite and it can be further decomposed into sub-activities
 - may be elementary and it is referred to as actions

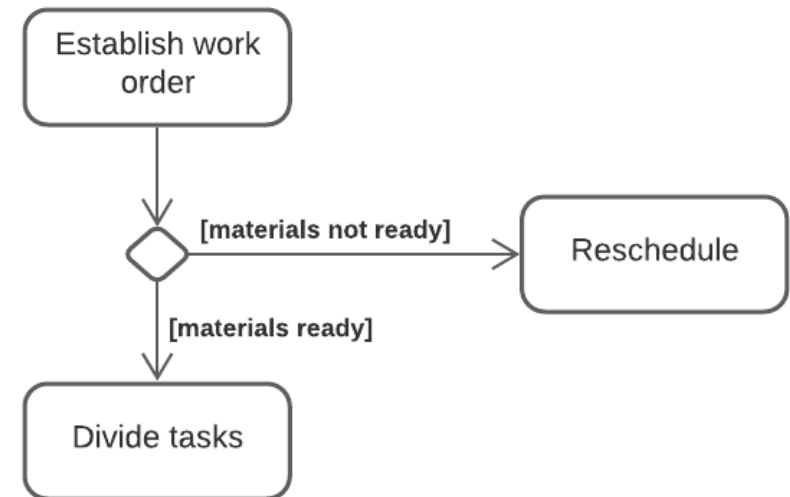
Activity diagram (cont.)



Activity diagram (cont.)

Branching

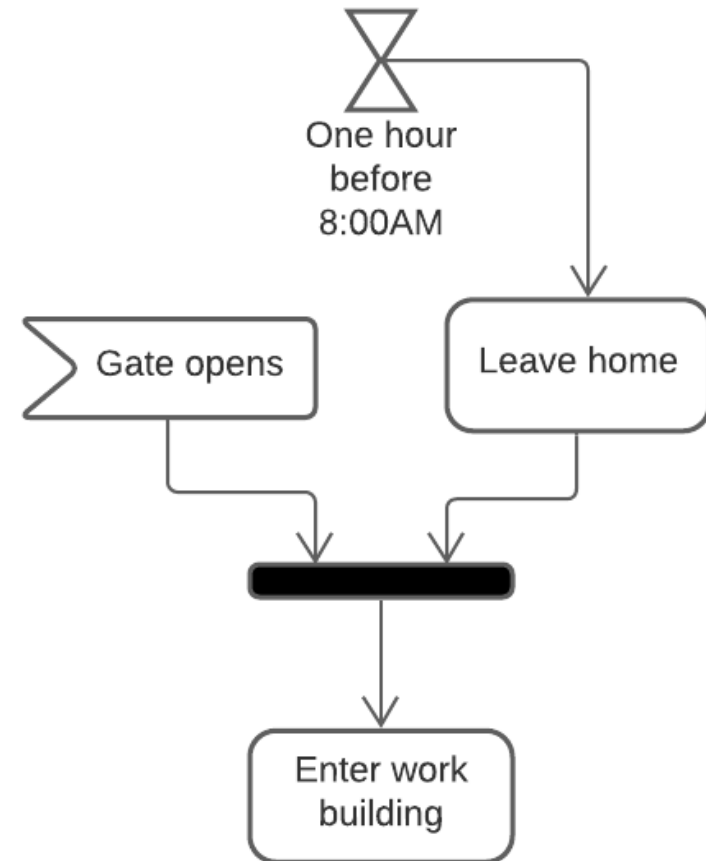
- Activities may be joined by a simple flow (arrow)
 - the completion of one activity leads immediately to the commencement of the other
- Activities may have a branching flow of control
 - i.e. some condition determines which is the next activity



Activity diagram (cont.)

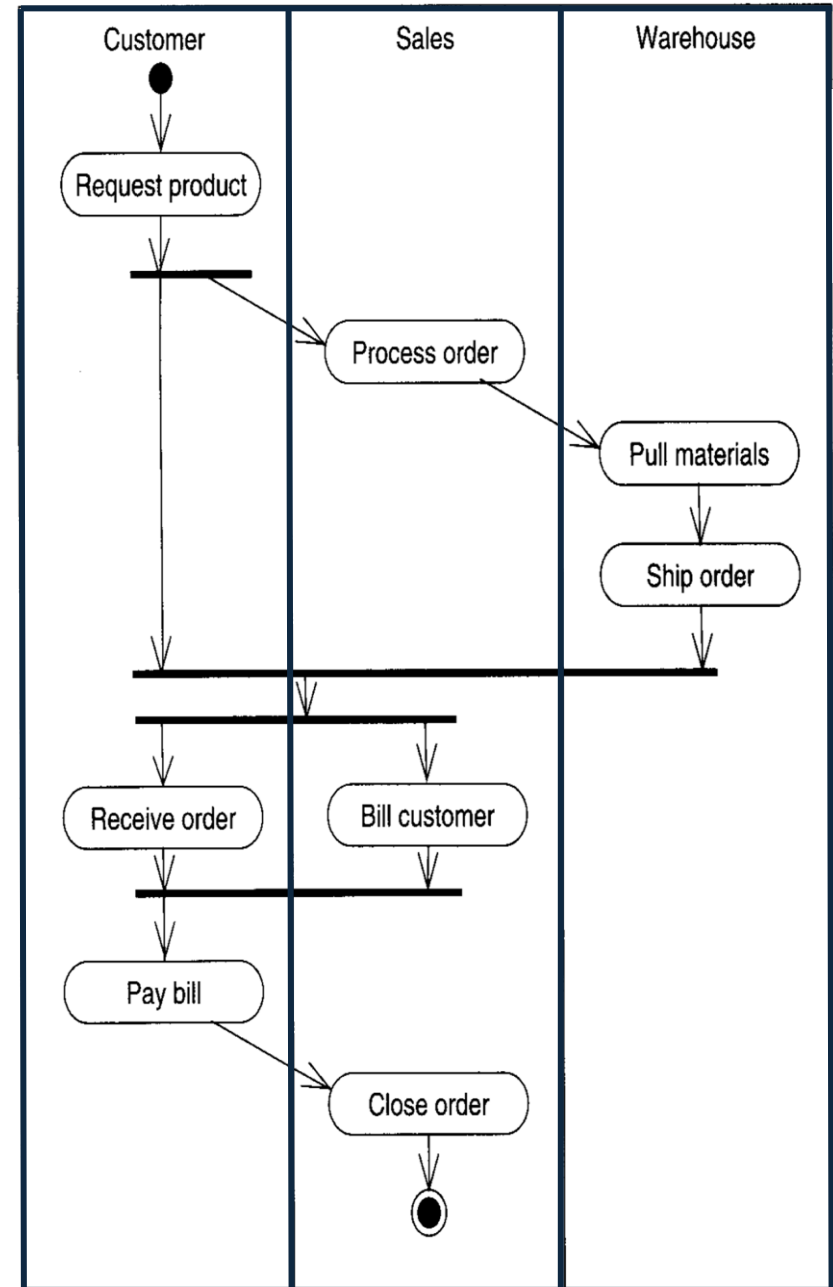
Forking and joining

- The commencement or completion of activities may need to be synchronised with other activities
 - e.g. some tasks in a project may not be able to commence till a number of others have completed



Swimlanes

- Allow engineers to model activities of a process for actors involved or related.



Activity diagram exercise 1

Context: a member is allowed to pick two services when arrives at the library: borrow or return. Then waits in the library queue. Upon call, the librarian help the member. In case of returning, the librarian record the returning and l put back the book on the shelf. In case of borrowing, the librarian record the borrow. By that, the service is done to the current member and librarian now can call for next member.

Draw an activity diagram that describes the activity above in two perspectives: member and librarian.

Activity diagram exercise 2

Context: an oven microwave heats or defrost items while its door is closed and the start button is pressed. A user needs to select/choose desire type and duration of service. Based on the selection the microwave starts working and displays a countdown on the screen.

Draw an activity diagram that describes the activity above in one perspective: user.

Activity diagram exercise 3

Context: you are working in a shipment company and you have some duties to handle once an order submitted to you by a customer. On receiving, you must send a confirmation of receiving the order to the customer and start filling the order details. There are two types of orders: 1) priority (must be delivered with 24 hours). 2) standard and must be put into a delivery queue.

Draw an activity diagram that shows the above scenario within two perspectives, the customer and the shipment company.